Lakshya Gupta

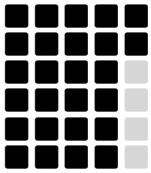
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Summary

Software Engineering and Computer Science undergraduate with a strong foundation in Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, and Computer Networks. Proficient in Python and Java with experience building ML-driven applications, automation tools, and scalable pipelines. Familiar with Git-based collaboration, REST APIs, Docker, and cloud fundamentals. Seeking a Summer Internship in Software Engineering to build reliable and high-performance systems.

Technical Skills

Languages: Python, Java, JavaScript, HTML, CSS, XML
CS Fundamentals: DSA, OOP, OS, DBMS, Computer Networks
Tools: Git, GitHub, Postman, AWS (Basics), Docker (Basics)
Libraries: TensorFlow, Transformers, Pandas, SciPy
ML Tools: LightGBM, XGBoost, Tesseract OCR
Concepts: REST APIs, Debugging, Testing, SOLID (Basics)



Projects

Birdie (Image Classification System)
Python, TensorFlow, ResNet-50, Pandas

- Developed an image classification system using transfer learning (ResNet-50) to classify 450 North American bird species.
- Built an end-to-end pipeline including data loading, preprocessing, augmentation, and training workflow automation.
- Achieved 94.1% accuracy after 50 epochs.
- Performed iterative evaluation and tuning to improve model generalization and classification accuracy.

Fuel Blend Properties (Ensemble Regression)
Python, LightGBM, XGBoost, Pandas, SciPy

- Built ensemble regression models using multi-output learning and per-target stacking to capture target correlations.
- Implemented per-target feature selection to improve prediction quality and reduce MAPE by 5% vs baseline.
- Achieved an additional 2% MAPE improvement through hybrid ensemble blending and hyperparameter tuning.
- Achieved 87.8% accuracy.

AI Voice Assistant (Automation Tool)
Python, SpeechRecognition, pyttsx3, Wikipedia API

- Designed a voice assistant with real-time speech-to-text (STT) and text-to-speech (TTS) for hands-free interaction.
- Integrated external APIs (Wikipedia, datetime) to handle natural language queries and return relevant results.
- Implemented modular command handling for browser automation, system tasks, and workflow execution.
- Improved stability using structured error handling, debugging, and input validation.

Education

Vellore Institute of Technology 2023 – 2027
B.Tech in Computer Science and Engineering

Jayshree Periwal High School 2022
Science (PCM)

Achievements & Activities

- **Competitive Programming:** Solved DSA problems on platforms like LeetCode / Codeforces.
- **Japanese:** JLPT N4 (Expected July 2026)
- **French:** Conversational Fluency